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Authors:

Veronica Lucchetti
Giovanni Saiu

Contents

1 Abstract

This project presents an implementation of a Single-factor Hull-White Extended Vasicek (1994) model for the dynamics of interest rates. The main purpose of this work is to analyze the investment of buying on May 26^{th}, 2016 a USD zero-coupon bond with maturity 18 months and sell it six months later. Two different approaches will be implemented: risk-neutral and real-world. The former will analyze the investment using the quotes of interbank deposits, interest rate swaps and the Black volatilities of ATM swaptions on the same IRS. While the latter will take into consideration the time series of the 6m Libor, 1y IRS, 2y IRS rate published by the Federal Reserve Bank of St. Louis between 2000 and 2015.

2 Literature Review

The Hull and White extension of the Vasicek model is one of the historically most important interest-rate models, being still nowadays used for risk-management purposes. Hull and White introduced a time-varying parameter in the Vasicek model, as they were trying to find an exact fit to the currently-observed yield curve. As a matter of fact, trying to match the model and the market term structures of rates at the current time is equivalent to solving a system with an infinite number of equation, one for each possible maturity. This system can be solved in general introducing a deterministic function of time, which we will call θ. Hull and White extension of the Vasicek model assumes that the short-rate process evolves under the risk-neutral measure according to[1]:

$$dr(t) = [\theta(t) - a \cdot r(t)] \cdot dt + \sigma dW(t) \tag{1}$$

where a and σ are positive constants and θ is a deterministic function of time. θ is chosen such that it fits the term structure of interest rates being currently observed in the market. If we denote the market instantaneous forward rate at time 0 for the maturity T by $f^M(0,T)$, *i.e.*

$$f^M(0,T) = -\frac{\partial \ln(P^M(0,T))}{\partial T} \tag{2}$$

where P^M (0, T) represents the market discount factor for the maturity. It can be proven that θ (t) must be:

$$\theta(t) = \frac{\partial f^M(0,t)}{\partial T} + af^M(0,t) + \frac{\sigma^2}{2a} \cdot (1 - e^{-2 \cdot a \cdot t}) \tag{3}$$

Therefore we can obtain the short-rate equation as:

$$\begin{aligned} r(t) &= r(s) \cdot e^{-a(t-s)} + \int_s^t (e^{-a(t-s)} \cdot \theta(u))du + \sigma \int_s^t (e^{-a(t-u)})dW(u) \\ &= r(s) \cdot e^{-a(t-s)} + \alpha(t) - \alpha(s)e^{-a(t-s)} + \sigma \cdot \int_s^t (e^{-a(t-u)})dW(u) \end{aligned} \tag{4}$$

where

$$\alpha(t) = f^M(0,t) + \frac{\sigma^2}{2 \cdot a^2} \cdot (1 - e^{-a \cdot t})^2 \tag{5}$$

As a consequence, r(t) conditional on $\mathcal{F}_s$ has Gaussian distribution with mean and variance:

$$E\{r(t)|\mathcal{F}_s\} = r(t) \cdot e^{-a \cdot (t-s)} + \alpha(s) \cdot e^{a \cdot (t-s)} \tag{6}$$

$$Var\{r(t)|\mathcal{F}_s\} = \frac{\sigma^2}{2 \cdot a} \cdot [1 - e^{-2 \cdot a(t-s)}] \tag{7}$$

At this point, the price at time t of a pure discount bond paying off 1 at time T is given by:

$$P(t,T) = E_t\{e^{-\int_t^T r(s)ds} H_T\} \tag{8}$$

Since, r(T) conditional on $\mathcal{F}_t$, $t \leq T$, has Gaussian distribution, also its integral form $\int_t^T r(u)du$ will have a normal distribution. So, we obtain:

$$P(t,T) = A(t,T) \cdot e^{-B(t,T) \cdot r(t)} \tag{9}$$

where

$$A(t,T) = \frac{P^M(0,T)}{P^M(0,t)} exp\{B(t,T) f^M(0,t) - \frac{\sigma^2}{4a}(1 - e^{-2at}) B(t,T)^2\} \tag{10}$$

$$B(t,T) = \frac{1}{a}[1 - e^{-a(T-t)}] \tag{11}$$

3 Point I: Risk-neutral Approach

3.1 Bootstrap

Firstly, we build the *bootstrap* function which implements the bootstrap of the Zero-Coupon curve from Money Market Deposits and IRS quotes (MID price) which are given in the excel spreadsheet. The discount for depos is computed using the following formula:

$$B(t_0, t_i) = \frac{1}{1 + \delta(t_0, t_i) \cdot L(t_0, t_i)} \tag{12}$$

Then, we compute the swap discount:

$$B(t_0, t_i) = \frac{1 - S_{IR}(0; t_i) \sum_{n=1}^{i-1} \delta(t_{n-1}, t_n) B(0, t_n)}{1 + \delta(t_{i-1}, t_i) \cdot S_{IR}(0; t_i)} \tag{13}$$

The function returns the bootstrapped dates and discount as vectors.
Then, we compute the zero rates using the function *zeroRates* which exploits the following formula:

$$y = -\frac{ln(B(t_0, t))}{\delta(t_0, t)} \tag{14}$$

In *Figure 1* we plot the results that we obtain from this computation.

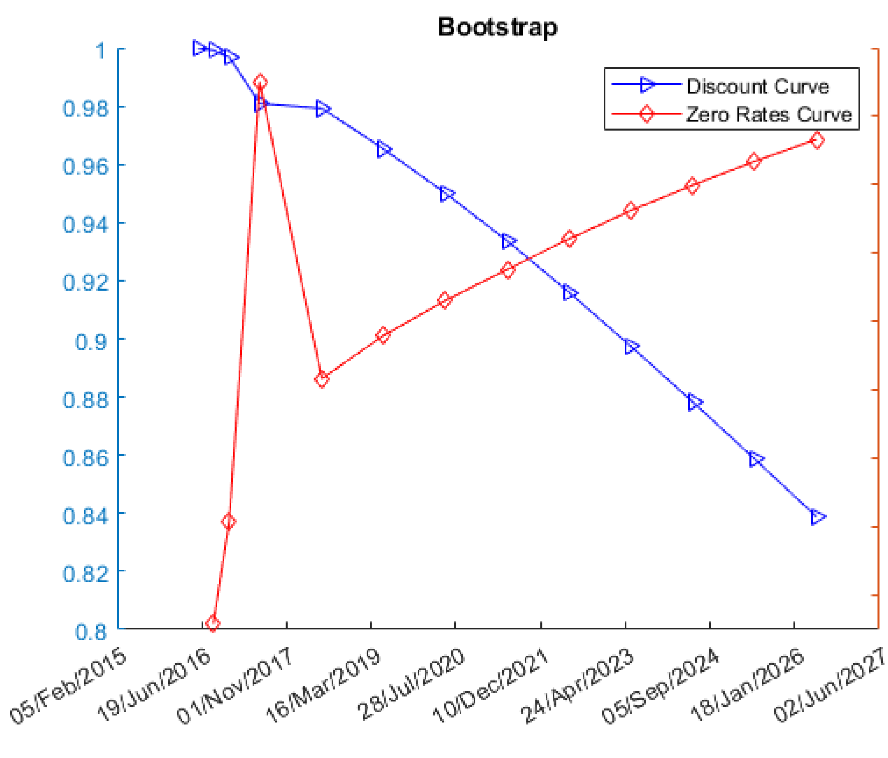

Figure 1: This image shows the discount curve and the Zero Rate Curve

3.2 Calibration of the Single-factor Hull-White Extended Vasicek (1994) model

At this point, we have to calibrate the Single-factor Hull-White Extended Vasicek (1994) model to the options with maturity 6 months and all underlying IRS maturities, in order to find the parameters a and *sigma*, minimizing the L2 price-distance.
We implement the function *Calibrate_HW_EV*.
First of all, the function computes the swaption strike using the function *Swaption_strike_ATM* which requires a struct with dates and discounts coming from the bootstrap.
Then, the interest-rate curve object is built from dates and data and therefore the annualized zero rate term structure.
At this point, we import the ATM Swaptions Black Volatilities (Mid Rate) with respect to the option which has 6 months of maturity. We then find the swaptions that expire on or before the maturity date of the sample swaption.
In order to calibrate the model minimizing the L2 price-distance we need to compute the observed market prices and predict the model's prices.
Since we are calibrating in order to find a and *sigma*, we are going to use function handles in which *param(1)=a* and *param(2)=* σ relative to our 6 months reference date.

The observed market prices are computed with the Black's Model. To do such computation we exploit the matlab function *swaptionbyblk* which prices swaptions using the Black option pricing model.
On the contrary, to find the model's predicted prices we price the swaption from a Hull-White interest rate tree. To price a swaption from a Hull White interest rate tree, we use the Matlab function *swaptionbyhw*. This function requires a Hull White interest rate tree, the values of the swap strike rate, Bond exercise dates and the Maturity date for each swap which we have all computed previously.
At this point, to minimize the L2 price-distance between the prices via Black Model and prices via Tree approach we exploit the Matlab function *lsqnonlin* which solves non-linear least squares problems. In doing this minimization we set as lower bound the value zero, since by hypothesis $a \geqslant 0$ and $\sigma \geqslant 0$, and as upper bound 1, since they are percentage values. We also set two starting values from which the minimization should start: $a = 0.01$ and $\sigma = 0.05$.
We obtain the following results, which are coherent:

Mean Reversion Rate (a)	Annual standard deviation of the short rate (σ)
0.4118	0.0101

Table 1

The following plot (*Figure 2*) represents the evolution of the zero curve of Hull White Model. The 3D graph axis represent the 20 semesters that we have in 10 years, the respective dates from 2016 to 2026 and a struct previously computed which contains the properties of an interest rate structure.

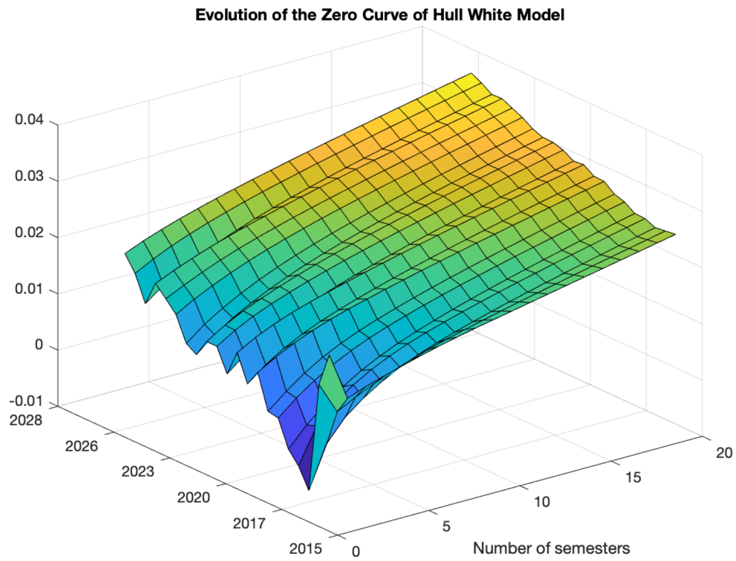

Figure 2

3.3 Risk-Neutral distribution of the zero coupon bond's prices 6 months later

First of all, we have to compute $B(t_0; t_0 + 0.50, t_0 + 1.50)$ which is an asset on which we do not have uncertainty, since it is an asset which starts at t_0. It can be computed with the following formula:

$$B(t_0; t_0 + 0.50, t_0 + 1.50) = \frac{B(t_0, t_0 + 1.50)}{B(t_0, t_0 + 0.50)} \tag{15}$$

In order to find $B(t_0, t_0 + 1.50)$ we need to interpolate the zero rate, in order to find the zero rate at 18 months (1.5 years) and then compute the discount value again.

At this point, we implement the *HW_EV_Forward_Bond* function which computes $B(t = 0.50; t, t + 1)$ exploiting the corollary of Lemma 1 in the following way:

$$\begin{aligned} B(t = 0.50; t, t+1) = B(t_0; T = 0.50, T + (\theta = 1)) \cdot exp\{-\frac{x_{0.50}}{\sigma} \cdot \\ \cdot (\sigma(0.50, T + \theta) - \sigma(0.50, 0.50)) - \frac{1}{2}\int_{t_0}^{T=0.5} (-\sigma^2(t, T + \theta) + \sigma^2(t, T))dt\} \end{aligned} \tag{16}$$

In order to compute $x_{0.50}$ we have simulated it using the Monte-Carlo simulation.
As result from these calculations, we obtain a vector of 500 elements whose mean, 95% confidence interval and standard deviation are equal to:

$B(t = 0.50; t, t+1)$	Expected Value	Confidence Interval	Standard Deviation
	0.9849	[0.98472 , 0.98499]	0.0016

Table 2

4 Point II: Investment Return

At this point, we assume to be an investor in fixed-income instruments. The investment we consider is to buy on May 26^{th} 2016 a USD zero-coupon bond with maturity 18 months and to sell it six months later.
Therefore, in order to calculate the risk neutral expected percentage return of this investment we consider it as if we at first buy a 18 months zero coupon bond and we sell it six months later. From previous calculation we already have the distribution of $B(t = 0.5, Ta = 0.5, Ti = 1.5)$ which we assume to be the price at which the zero coupon bond will be sold in 6 months.
At this point, we are left with the calculation of the price of a zero coupon bond issued today with maturity 18 months. In order to compute this price, we use the Single-factor Hull-White Extended Vasicek (1994) model because its advantage is that it has a direct formula for the price of simple financial instruments such as zero-coupon bonds. Indeed, the price at time t of a pure discount bond paying off 1 at time T is given by [1] :

$$P(t,T) = A(t,T)e^{-B(t,T)r(t)} \tag{17}$$

where functions A and B are not stochastic and are fully determined by the model parameters. In the Hull-White extended Vasicek model:

$$A(t,T) = \frac{P^M(0,T)}{P^M(0,t)} exp\{B(t,T)f^M(0,t) - \frac{\sigma^2}{4a}(1 - e^{-2at})B(t,T)^2\} \tag{18}$$

$$B(t,T) = \frac{1}{a}[1 - e^{-a(T-t)}] \tag{19}$$

where all the variables are computed according to the equations explained in the Literature Review section.
In our case t=0 is May 26^{th}, 2016 and T=18 months is the maturity.
We implement the *Affine_trick* function in order to derive A(t,T) and B(t,T).
Firstly, this function computes the instantaneous forward rate $f(0,t)$ through numerical derivative, using the increment ratio. Then, it computes B(t,T) and A(t,T) according to their formulas.

We obtain the following value of the price at time $t_0 = 0$ of the zero coupon bond with maturity T=18m:

$$P(0, 18m) = 0.9815 \tag{20}$$

In order to compute the expected percentage return and the standard deviation of the investment we exploit the following formula:

$$Rate\_of\_return = \frac{Selling\_value - Initial\_value}{Initial\_Value} \cdot 100 \tag{21}$$

We obtain the following results:

Expected return rate of the distribution	0.34092 %
95% confidence interval of the distribution	[0.32687,0.35497]
Standard deviation rate of the distribution	0.15993

Table 3

From these results we can suppose that the investment is expected to return the 0.34 % of its value. In the following image (*Figure 3*) we plot:

- the risk-neutral distribution 6 months later of the zero coupon bonds' prices $B(t = 0.50; t, t+1)$ which has 12 months left of maturity
- the expected value of $B(t = 0.50; t, t+1)$
- the zero-coupon bonds' price $B(t_0; t_0 + 0.50, t_0 + 1.50)$

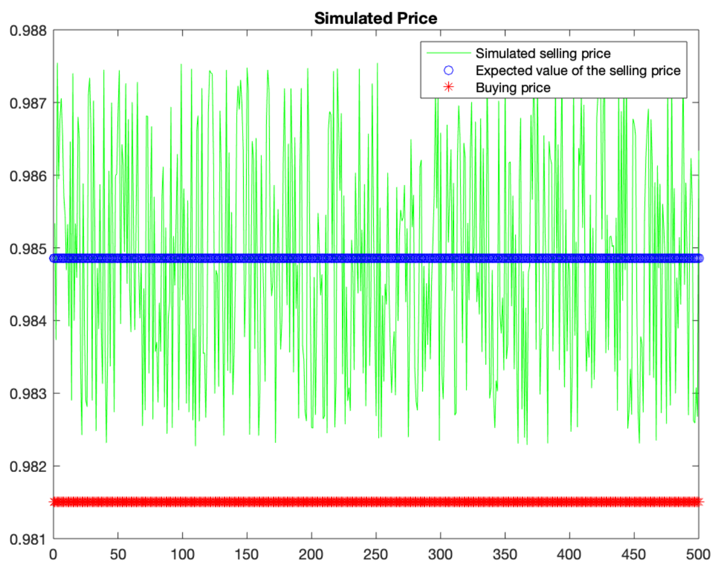

Figure 3: Distribution of the selling price $B(t = 0.50; t, t+1)$, its expected value and the buying price $B(t_0; t_0 + 0.50, t_0 + 1.50)$

Indeed, as the image shows, the expected value of the zero-coupon bonds' prices $B(t = 0.50; t, t+1)$ 6 months later with 1 year left of maturity is higher than the original one with 1 year and a half of maturity $B(t_0; t_0 + 0.50, t_0 + 1.50)$.
Hence, we can conclude that the investment has a positive expected percentage return.
As a matter of fact, in the following image (*Figure 4*) we plot the distribution of the investment return, which is positive, and we also plot its expected value which is equal to 0.34092 %.

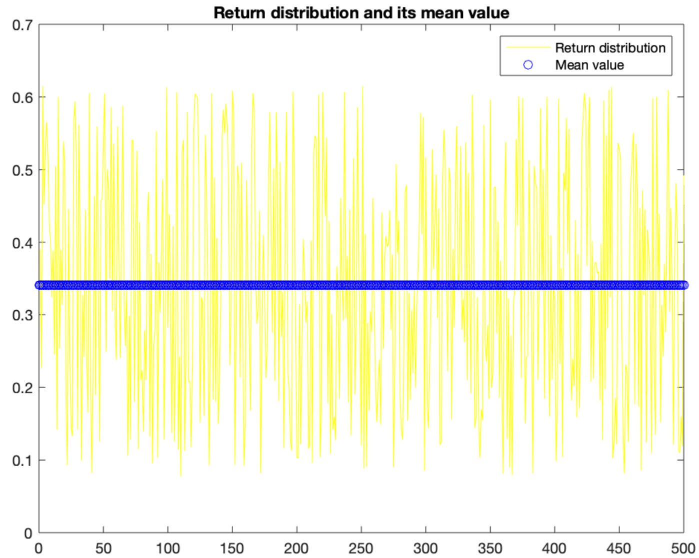

Figure 4: Distribution of the percentage return of the investment and its expected value

5 Point III: Real World Approach

In this part, we are going to use the time series of the 6m Libor, 1y IRS, 2y IRS rate since year 2000. These time series are published by the Federal Reserve Bank of St. Louis. We use the given time series in order to derive the time series of the zero-coupon curves (defined on three nodes: 6m, 1y, 2y) at semiannual frequency.
We take into consideration the last working day of the semester from the first half of 2001 to the second half of 2015.
Then, based on these time series of the zero-coupon curves, we will evaluate the historical (real-word) distribution of realized percentage return of the investment into a zero-coupon bond with 18 months original maturity sold six months later.

5.1 Zero-Coupon Bond Pricing

Firstly, we import only business dates every 6 months from 2001 to 2015. To do this implementation we exploit the function *dateMoveVec*. This function moves a given date forward or backward by a given number of time units according to a given business days convention on a given market. Therefore, we consider as settlement date December 31^{st}, 2000 and we move forward of 30 semesters (Remark: from 2001 to 2015) .

At this point, we create a matrix which contains:

- in the first column the vector of the last working day of each semester
- in the second column we import from the Excel Spreedsheet for each of our 30 dates the corresponding 6-Month London Interbank Offered Rate (6m LIBOR)
- in the third and fifth column we import from the Excel Spreedsheet for each of our 30 dates the corresponding 1-Year and 2-Year Swap Rate respectively.

At this point, we have a matrix with only the semiannual, annual and biannual rates. However, our bond has a 18 months maturity. Therefore, we implement a linear interpolation for each row of the matrix in order to get the 18 months historical rate. Then, we insert it into the fourth column of the matrix.
Subsequently, we derive from the historical distribution the rates we expect to have on May 26^{th}, 2016, which is the day on which we buy the zero-coupon bond and on November 25^{th}, 2016 , which is when we sell it. This computation is done through a linear extrapolation technique.

Then, we derive from the data the zero-coupon curve exploiting the *ZC_bootstrap_IRS_only* function. This function returns a matrix containing expires and zero rates for the swap rates given as input in a matrix form. Firstly, the function takes the 1y, 1.5y and 2y interest rate swap and computes the relative discount factors, with the following formula:

$$B(t_0, t_i) = \frac{1 - S_{IR}(0; t_i) \sum_{n=1}^{i-1} \delta(t_{n-1}, t_n) B(0, t_n)}{1 + \delta(t_{i-1}, t_i) \cdot S_{IR}(0; t_i)} \tag{22}$$

Then, it computes the zero rates as follows:

$$y = -\frac{log(B(t_0, t))}{\delta(t_0, t)} \tag{23}$$

Finally, we derive from this curve the prices of the zero-coupon bond using the following formula[3]:

$$P(t, T) = E_t[e^{-\int_t^T r_s ds} \cdot 1] \tag{24}$$

where we take as r the zero rate previously calculated.

More precisely, according to the previous formula, we calculate how much it would have cost to buy a zero-coupon bond with 18 months of maturity at each semester and we will have a curve of historical prices. Then, we assume that when we will sell the zero-coupon bond six months later, it will have 1 year of maturity left, so we consider how much it would have cost to buy a 1 year maturity zero-coupon bond during those years and we obtain our second curve of prices.
The two curve of prices have the following expected value and standard deviation.

Bond	Expected Price	Standard Deviation
18 months of maturity left	0.9794	0.0164
12 months of maturity left	0.9903	0.0086

Table 4: Expected value and standard deviation of the values of the two bond with respectively 18 and 12 months left of maturity

As the table shows, it is easy to deduce that we will have a positive return.
The following image (*Figure 5*) shows a comparison between the trajectory of the prices of a zero-coupon bond with 18 months of maturity and the prices of a zero-coupon bond with 12 months left of maturity evaluated at the following semester.

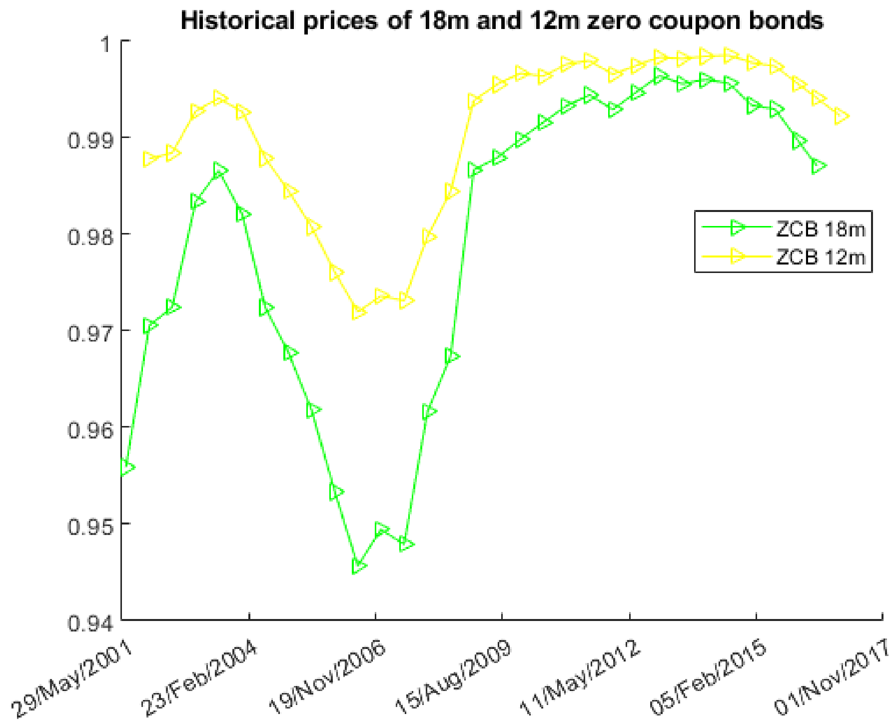

Figure 5: Trajectory of the prices of a zero-coupon bond 18m with the ones of a zero-coupon bond 12m valuated at the following semester

5.2 Return on the investment

To derive the distribution of realized percentage return of the investment into a zero-coupon bond with 18 months original maturity sold six months later, we exploit the formula

$$Rate\_of\_return = \frac{Selling\_value - Initial\_value}{Initial\_Value} \cdot 100 \tag{25}$$

Where

- Initial value is the price of the zero-coupon bond with 18 months of maturity, which is computed at our settlement date (May 26^{th}, 2016) and it is derived from the historical series.
- For selling value we assume that the zero coupon bond will be bought at the price at which the 12 months zero-coupon bond maturity are being sold.

As far as the selling price is concerned we consider two different approaches.
The first approach is to take as selling price the whole historical distribution of prices of the zero-coupon bonds with 12 months of maturity. That because we assume it would be a more precise comparison than only taking the extrapolated price at November 25^{th},2016. At this point, we can compute the distribution of the rate of return according to *Equation 25*. We obtain the following results for the expected return rate and standard deviation rate:

Expected return rate of our investment	Standard deviation rate of our investment
0.32711	0.86914

Table 5: Expected return rate of our investment and standard deviation - Real World Approach

Hence, we can deduce that the investment is expected to return the 0.32 % of its value.

Secondly, we consider also as selling price only the extrapolated price at November 25^{th},2016. Therefore, we will not have a distribution but a single value which is:

Rate of return of our investment
0.52481

Table 6: Return rate derived from extrapolated prices only

Hence, the investment is expected to return the 0.52 % of its value, which is higher than the previous case and also higher than the expected return computed through the risk neutral approach.

Regarding the first approach the following plot (*Figure 6*) illustrates the distribution of the return on our investment and its expected value. As before, we expect to have a positive return on dates close to our settlement date. However, it is normal to observe also negative returns because we are making a comparison between a buying price estimated at May 26^{th}, 2016 and a selling price of previous years. Especially, we notice that we have a negative return in years close to 2008 which is when the market suffered a global financial crisis.

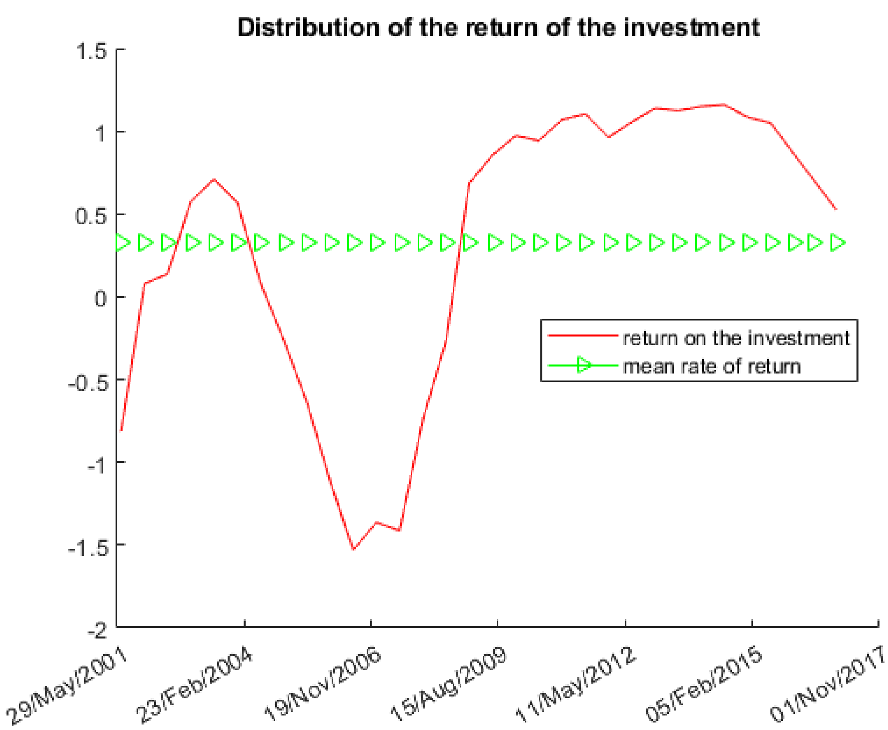

Figure 6: Distribution of the return of the investment and its expected value

6 Point IV: Comparison between the risk-neutral and real-world distributions

In the previous points we have faced the same problem with two different points of view, risk-neutral and real-world:

- In the risk-neutral case (point I and II) we are given a market data with many nodes to bootstrap: 2 Money Market Deposits (3m and 6m) and 10 Interest Rate Swaps (1y, 2y, 3y, 4y, 5y, 6y, 7y, 8y, 9y, 10y). We do a risk-neutral calibration and we get a distribution of the possible values of the two financial instruments: the bond with 18 months of maturity and the bond with 12 months of maturity, but at 6 months from now. We consider the ZCB as an asset. More precisely, we buy now one asset which values x and we sell after 6 months one asset which values y, where y is stochastic. From the distribution of y we can derive the distribution of the return of our investment.
- In the real-world (point III) we do the same thing, though considering historical values. We have from the Excel spreadsheet 3 financial instruments (6m Libor, 1y Swap Rate, 2y Swap Rate) which we use to compute ZC curve defined only on 3 expiries. We obtain the expiry at 18 months with an interpolation. Then we compute the bond's prices exploiting the outputs of the bootstrap.

From the two methodologies we obtain the following results for Expected Percentage Return and its Standard Deviation:

	Expected Percentage Return	Standard Deviation
Risk-Neutral	0.34092	0.15993
Real-World	0.32711	0.86914

Table 7

As we can see, the two distributions have a similar expected value but different standard deviation.
We can explain this difference with the fact that in the first case we obtain an analytical distribution, while in the second one it is a sample distribution. It is the same problem seen from two different points of view. The first two points belong to the pricing, while the third is an econometric approach even though we have to implement the bootstrap.
Therefore, we can conclude that the two approaches lead to the same conclusions for the fixed-income investor. As a matter of fact, in both cases we obtain an expected percentage return of around 0.3% which means that the investor will gain from this investment and that the Single-factor Hull-White Extended Vasicek (1994) model offers a reliable estimation of the return. The standard deviation is higher when we face the real world problem probably because we based our analysis on a 15 years window which faced multiple ups and downs in the pricing market.

7 Point V: A different calibration approach

In this point we are required to repeat the calibration made in point I with input the ATM volatility of the options with maturity 3 years and all underlying IRS maturities (1y, 2y, 3y, 4y, 5y) and derive the risk-neutral distribution of the investment in the zero-coupon bond.
The computation is the same as point I, as a matter of fact when we create the function *Calibrate_HW_EV* we introduce a flag value to state the difference in using either the options with maturity 6 months or the options with maturity 3 years. Therefore, the only difference is that we import the ATM Swaptions Black Volatilities (Mid Rate) with respect to the option which has 3 years of maturity. In this case we obtain, as expected two different values of a and σ which are still coherent:

In input the ATM volatility of	Mean Reversion Rate (a)	Annual std of the short rate (σ)
Options with 6m maturity	0.4118	0.0101
Options with 3y maturity	0.012143	0.0092161

Table 8: This table shows the a and σ values calibrated with different ATM volatility in input

What can be noticed is that both values decrease of one order of magnitude. However, a decreases significantly much more than σ. As we know a represents the rate of the mean reversion which is the process that describes that when the short-rate r is high, it will tend to be pulled back towards the long-term average level; when the rate is low, it will have an upward drift towards the average level. Having a smaller value of this rate might mean that the short rate does not have to be pulled back as quickly as before and therefore the rate is more static toward its long-term average mean value.

The following plot (*Figure 7*) represents the evolution of the zero curve of Hull White Model. The 3D graph axis represent the 10 years, the respective dates from 2016 to 2026 and a struct previously computed which contains the properties of an interest rate structure.

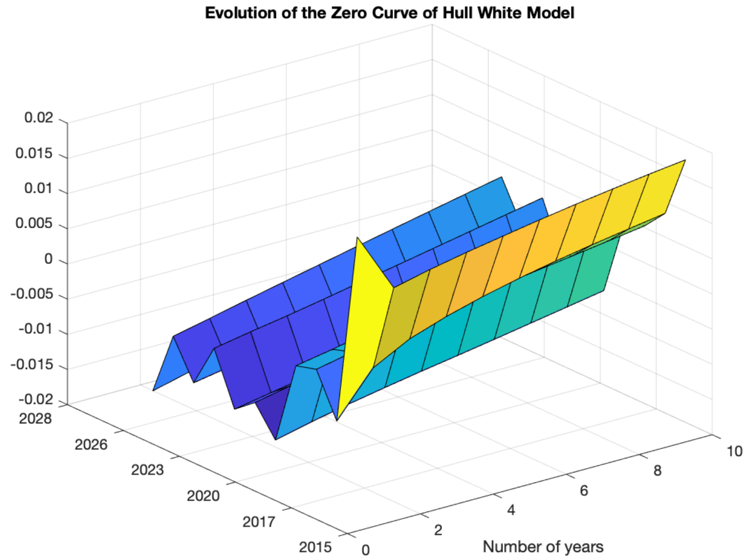

Figure 7

At this point, we can repeat what we did for point I and II. Firstly, we compute $B(t_0; t_0 + 0.50, t_0 + 1.50)$ exploiting as before *Equation 15*. Then, we compute $B(t = 0.50; t, t + 1)$ using the *HW_EV_Forward_Bond* function which exploits the corollary of Lemma 1.
We obtain the following results:

$B(t = 0.50; t, t + 1)$	Expected Value	95% Confidence Interval	Standard Deviation
	0.98425	[0.98408,0.98442]	0.0019234

Table 9

We obtain the following value of the price at time $t_0 = 0$ of the zero coupon bond with maturity T=18m:

$$P(0, 18m) = 0.98261 \tag{26}$$

If we compute the expected percentage return using the same formula of point II (*Formula 21*) we find that:

Expected return rate of the distribution	0.16636 %
95% confidence interval of the distribution	[0.14916 , 0.18355]
Standard deviation rate of the distribution	0.19574

Table 10

By comparing these results to the ones obtain in point II (*Table 3*) we notice that the standard deviation is higher than before (0.19 compared to 0.16), while the expected return is lower (0.16 % compared to 0.34 %).
This is probably due to the fact that we are considering zero-coupon bonds with maximum maturity of 18 months and calibrating the model referring to the volatilities of the options with

maturity 3 years will provide us with less precision on the estimation and more uncertainty on our rate of return.
In the following image (*Figure 8*) we plot as before : the risk-neutral distribution 6 months later of the zero coupon bonds' prices $B(t = 0.50; t, t + 1)$, its expected value and the zero-coupon bonds' price $B(t_0; t_0 + 0.50, t_0 + 1.50)$.

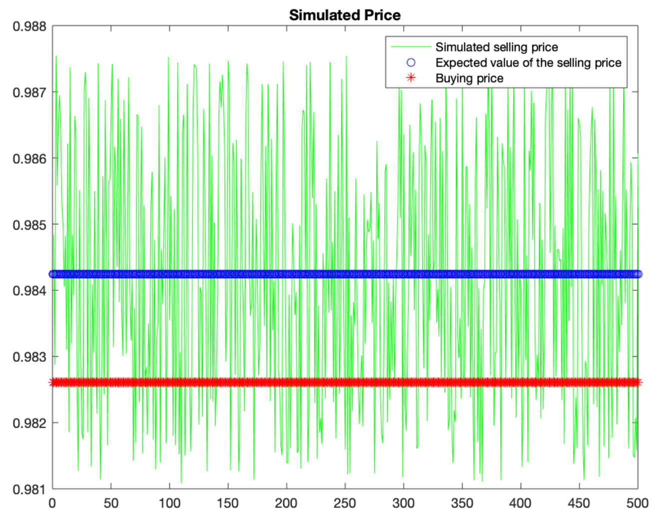

Figure 8: Distribution of the selling price $B(t = 0.50; t, t+1)$, its expected value and the buying price $B(t_0; t_0 + 0.50, t_0 + 1.50)$

Even in this case we can notice that the expected value of the zero-coupon bonds' prices $B(t = 0.50; t, t + 1)$ 6 months later with 1 year left of maturity is higher than the original one with 1 year and a half of maturity $B(t_0; t_0 + 0.50, t_0 + 1.50)$. Hence, we can conclude that the investment has a positive expected percentage return.
However, if we compare *Figure 8* with *Figure 3*, we can highlight that in the first case the simulated selling price trajectory is always above the buying price value. On the contrary, in this case the simulated selling price fluctuates more and it touches the buying price. In addition, in this case the simulated selling price values are included between [0.9811 , 0.9875] while in the previous case between [0.9823 , 0.9875]. This due do the higher standard deviation that we have in this case.

In the following image (*Figure 9*) we plot the distribution of the investment return:

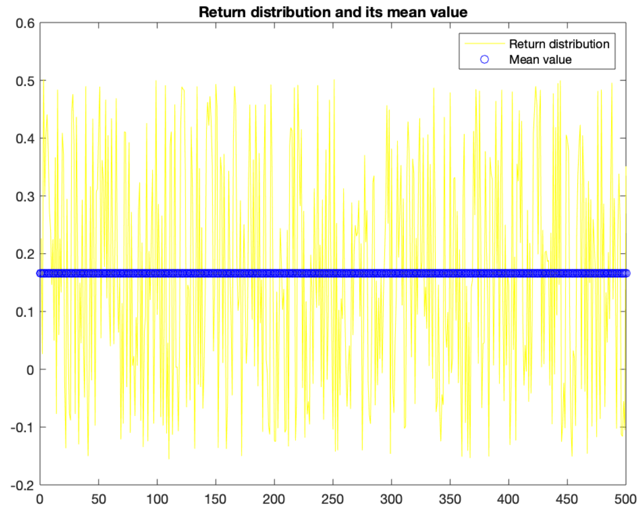

Figure 9: Distribution of the percentage return of the investment and its expected value

As expected, we get a lower expected percentage return (0.16 % compared to 0.34 %). That is because in *Figure 8* sometimes the simulated selling price is below the buying price and therefore the return distribution takes also negative values, which did not happen before. However, the mean value is still positive.

In conclusion, we can state that the best choice for the fixed-income investor is to calibrate the Single-factor Hull-White Extended Vasicek model to the options with maturity 6 months. As a matter of fact, this calibration choice leads to a higher expected percentage return with a smaller standard deviation.

8 Bibliography

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